

AI Powered OCR Solution for Tamil Historical Document

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Abstract—Palm leaves, plates of copper and stone inscriptions which were used in ancient Tamil are an invaluable cultural source yet manual reading of these sources is largely hampered by the physical decay, the blurring of the scripts and complicated ligatures. Existing optical character recognition (OCR) technologies cannot recognize such historical documents well due to noise, variation in spacing and changes in style. This project presents an artificial intelligence-based OCR system that is specifically trained to work with ancient Tamil manuscripts. Through a semi-supervised form of deep learning, the system combines a convolutional neural network (CNN) (ResNet50) (spatial feature extractor) and a bidirectional long short-term memory (BiLSTM) layer to efficiently acquire sequential dependencies. Additionally, data augmentation and pseudo-labeling are applied to address the issue of annotated datasets scarcity. The system suggested takes images and preprocesses them systematically extracting hierarchical features, modeling character relationship in context and generating Unicode Tamil text hence making it editable and searchable. This work is innovative as it combines semi-supervised learning with a hybrid CNN-BiLSTM model that is specially designed to take into account ancient Tamil scripts in poor condition thus allowing accurate digitalizing, long maintainability, and improved availability of historical information.

Keywords—Ancient Tamil, Palm Leaf Manuscripts, Historical Preservation, Digital Archiving, Script Recognition, Cultural Heritage, Degraded Texts.

I. INTRODUCTION

Tamil is considered as one of the oldest classical languages in the world, and public tradition of literature is more than two thousand years old. Its rich culture and history is preserved by inscriptions dated back to palm leaves, copper plates and stone inscriptions. The inscriptions provide priceless information about the governance of ancient societies, their trade, literature, medicine, philosophy and religious beliefs of the ancient societies. However, it is especially [1] the nature of the tangible materials onto which these texts are inscribed, which is especially vulnerable to progression of nature, as well as environmental factors and interventions of humans. Palm leaves are likely to break off, cracks are likely to occur and the leaves are also likely to be infested by insects or fungi, whereas the stones and copper inscriptions erode, fade and suffer surface damages over centuries. As a result, large amounts of historical knowledge are also at the threat of being lost forever.

The work of preserving these texts is not only an academic one but a very important measure towards cultural identity protection and guaranteeing continuity of the knowledge in

the new generations. Although the modern day technology can provide the solution in the form of digitisation which can translate the fragile manuscripts, it preserves the content and at the same time, can be accessed by researchers [2], teachers and the general audience. The core of the process is Optical Character Recognition (OCR) that converts handwriting, as well as engravings, programmatically into the digital formats.

Although OCR has greatly advanced to meet the requirements of the modern printed scripts, it suffers challenges that are immeasurable when it is applied to ancient scripts besides complex scripts such as the Tamil scripts. There are special challenges with irregular spacing between characters, strokes that fade or are missing, ligatures, stylistic variations, noisier backgrounds presented by inherent texture of palm leaves or stone, and so forth. All these difficulties lead to the ineffective work of the standard OCR engines [3], which is why special attention should be paid to approaches developed to work with ancient scripts. The artificial intelligence, especially deep learning, has demonstrated tremendous potential in addressing these challenges. Neural networks are able to capture complicated spatial patterns, identify sequential dependencies as well as generalize to several patterns of writing.

By integrating the hierarchical feature learning, a system that is guided by the context of sequence modeling, the AI system will be able to identify characters even in extreme levels of degradation. Moreover, traditional supervised learning algorithms cannot be applied on old datasets which are small in size, and which are not richly annotated. The scalable solution to the low-resource [4] optical character recognition (OCR) problem is semi-supervised techniques that exploit both labeled and unlabeled data and techniques to synthesize degradation of real-world data. The main goal is to develop a system that is capable of transforming Tamil ancient inscriptions into edible, searchable, and Unicode compatible digital text.

This reprocessing does not only contribute to preservation but also makes it easier to access, hence permitting historians, linguists, students at all parts of the world to play with knowledge that was centuries old. They can be used as a basis to carry out a linguistic study, research on how Tamil evolved in history, phonetics, and its syntax, through such digital archives. Schools can include such resources as part of their learning curriculum, whereas cultural institutions [5] can use the digitized archives to display exhibitions and engage with them. Besides the practical conservation, the automation of the identification of an ancient Tamil inscriptions can also contribute to the research in the field of artificial intelligence,

especially in those domains that are defined by the lack of annotated information and a large amount of variability in the quality of input information.

Creating a strong identification protocol to the use of historical scripts also creates possibilities to the cross-script adaptation, which can eventually improve the process of digitizing other South Indian and classical languages. This approach [6] will ensure that the linguistic and historical richness of the Tamil civilization is not only not lost in the digital age but also made highly accessible to ensure that interdisciplinary scholarship and enhancement of cultural continuity.

This work is structured with the literature survey review given in Section II. Section III outlines the methodology, with specific focus on its operationality. Results and discussions are in Section IV. Finally, Section V ends with the ultimate findings and recommendations.

II. LITERATURE SURVEY

The study of Optical Character Recognition (OCR) for ancient scripts has witnessed growing interest due to the dire need for cultural conservation and the digitization of ancient documents. Current academically available work has mostly focused on modern printed documents, providing few solutions for handwritten or faded manuscripts. Earlier approaches combining standard OCR engines with machine learning attained an average level of performance; however, they suffered when dealing with rich scripts, ligatures, and noisy inputs. Contemporary developments in deep learning using convolutional and recurrent neural networks have showcased promise in the identification of spatial as well as sequential patterns in the context of the Indo-Aryan and Dravidian language groups. This literature review critically reviews the past work on OCR methodologies, deep learning architectures, as well as semi-supervised methodologies relevant for ancient inscriptions in Tamil.

The ability to read handwritten text is still a challenging feature that is used in document analysis, postal or through historical preservation. The technology has enhanced the processing of offline and online scripts in various writing styles. The study draws attention to data preparation, feature identification [7] as well as optimization of recognition processes. Its comparison against standard data sets indicates its strengths and weaknesses and can serve to improve in future. The major challenges are mixed scripts, writing in cursives and loud inputs. Novel methods are still emphasizing on these issues providing a robust site to produce workable and precise recognition systems. The field presents crucial information to scholars and individuals trying to enhance handwritten text processing.

Digitization of historical documents will alter the tangible materials of archives into digital versions of the same item in order to preserve, retrieve, and interpolate the information. The process has the effect of increasing accessibility to scattered collections, thereby enabling effective data mining and academic research. Workflows can separate image acquisition, cleaning and transcription as well as deal with the issues of complicated layouts [8] and poor scripts. Assessments that are based on datasets highlight the weaknesses of existing methodologies and guide their refinement. Digitization of materials requires standardised procedures to guarantee integrity. The discipline also encourages cross-disciplinary research and provides

procedures to save historical material thus making digitization an inseparable element towards successful conservation, analysis, and sharing of historical information over the digital arena.

The high-quality text data construction is significant in the application of past or industry text data. It entails gathering, cleaning, labeling as well as enhancing heterogeneous and complex materials to render them trustworthy. Application-specific knowledge is useful in improving the quality of datasets [9], and reducing error in the analysis. Curation sets are useful in promoting accurate knowledge, effective research, and training models. The text collections of high quality are used in exploration of history, industrial studies and also systematic learning. Knowledge of experts can make the interpretability and usability unquestionable, and noise is minimized. Reliable datasets are used as foundation to effective text investigation and algorithmic calculations in order to make substantial assumptions concerning various and problematic textual information.

The OCR and HTR technology can preserve and bring the previously written historical documentation written in various languages and in various scribal traditions. When materials such as the Old Nubian materials are digitized, this enhances study, translation, and analysis and at the same time maintains delicate originals. Accurate recognition contributes to effective search and analysis of historical materials, hence contributing to scholarship in all regions of the world. The maintenance of various textual traditions depends on retention [10] of multilingual functions. Detailed recognition systems are also essential to the documents writing mixed scripts because they make the researcher study personnel history effectively. Digitization is also influential in cultural preservation and academia to ensure that valuable manuscripts are kept in digital form and accessible to be accessed and utilized by the future scholars without undermining the authentic content.

The recognition of rare scripts, like Syloti Nagri, is a matter of dealing with complex shapes of characters and ligatures. Narrow datasets need specialized procedure in order to achieve accuracy. Good recognition will aid in managing archives, research, and education by availing credible [11] access to the historical scripts. The preservation of lingual and cultural continuity is made by capturing textual information in detail. Endangered scripts can be studied and analyzed by means of digitization, which improves the preservation and sharing of knowledge. Proper depiction of the rare scripts enhances the culture and offers great resources to the scholars, students and the communities that would like to uphold historical linguistic traditions.

Handwritten text and printed text mixed documents are essential to transcription and analysis. Pixel techniques can be used to differentiate properly without initial assumptions of the layout. Image enhancement and segmentation enhance the clarity of small or intersecting texts. Effective separation facilitates layout study, identification and historical study. The continuance of textual [12] information makes the documentation of documents of varying structures trustworthy. This helps in digitalization of archives, learning and studies of manuscripts. Precise differentiation of text types allows efficient processing and preservation of the document integrity, thus making research and computational applications easily achievable besides being able to study the handwritings and printed materials further.

The digital collections of the past can be analyzed at large scale through cloud-based solutions. They help in information extraction, visualization and time research and thus facilitate research in different fields. With the use of parallel processing, one will be able to deal with high volumes of data and complex analyses. Such systems facilitates the discovery of historical resources [13], which unravels patterns and associations. The usability and creation of insights are enhanced with real-time processing and automated visualization. Availability of cloud technology promotes collaboration, scalability and reproducibility of historical research. These platforms harness the capacity of digitized archives to provide organized and interpretable forms of outputs that help researchers to successfully study and analyze complex textual collections of history at a global scale.

The conversion of ancient texts into contemporary multilingual versions will help to keep the cultural knowledge and to increase the accessibility. The computerization of the Sanskrit manuscripts makes retrieval, translation and making of audio possible in real time. Better readability and interpretability are beneficial to research, teaching and tourism. The availability of transformed works allows people to grasp the traditions, the historical background [14], and the development of language. It is crucial to promote authenticity by ensuring that the content is accurate. The digital technologies are used to bridge the gap between historical manuscripts and the current populations and thus to facilitate a wider accessibility. Such activities lead to preservation of culture, education and interactive sessions which make historical facts easier and also preserve the old works so that the future generation can use them as well as encourage old written materials to be relevant and be utilized in the contemporary context.

The fact that historical materials are preserved in rare scripts, e.g., Sino-Nom, would require the usage of special recognition systems and data pipelines. The transcription and digital availability of these scripts is easy owing to the collection and subsequent analysis of the scripts. Good datasets may result in recognition and detection, hence, historical research and cultural preservation. The digital encourages the scholarly study of books [15], politics, and social environments. These organized entry are the mechanisms that ensure that there is continuity in availability of the rare scriptures. Thorough strategies increase the level of recognition and help with management of the archives as they offer reliable means of analysis and interpretation. Moreover, with digitization, cultural memory is secured, and researchers are able to analyze past documentation conveniently, making sure that the useful documentation is available, understandable, and can be saved to be studied in the future.

Associative retrieval of historical writings, the concealed relationships among individuals, places, and times can be found. Along with the context analysis, semantic graphs identify both direct and indirect mentions across the varied sources. The resulting associations which are visualized help to see social structures and historical networks that are difficult to distinguish otherwise. This is a qualitative advantage of genealogy research [16], but also of cultural studies. The extraction of relations in damaged or noisy documents will allow reconstructing not only the interaction but the events as well. Formalized techniques are open to knowledge discovery, which gives social, political, even cultural background. This kind of processing of the historical

documents enhances comprehension of the complex networks, bringing into view associations that were not known without enhancing interpretation of the ancient societies.

In complex written books, including Ottoman scripts, segmenting the text lines is important to be recognized and analyzed. By comparing deep-learning strategies, one can identify effective patterns of dealing with ornate, overlapping, or cursive calligraphy. Detailed segmentation preserves the details, whereas instance recognition is provided with the help of detection-based methods. Assessments allow accurate transmission and archiving of [17]-lines. Strong segmentation complements the results of later recognition and analysis, which contributes to greater access to historical materials. The preservation, research and digitization activities are assisted by techniques that can serve various writing styles.

The dependable line extraction enables some considered processing performance on the complex manuscripts thus assisting in the efficient management and examination of historical texts with convoluted layouts. The identification of historical Ethiopic handwritten documents involves working with a large number of characters and diverse styles of writing. Sequential feature extraction end-to-end [18] strategies improve the accuracy of transcription. Data augmentation handles small datasets, enhancing the performance of models. Proper identification retains manuscripts and makes them digitized so that they can be used by scholars. These strategies enhance the availability of exclusive cultural information and facilitate linguistic research. There are credible tools of recognition that facilitate the preservation of heritage materials as they are being created.

The manuscripts in Ethiopic language, which could be studied and analyzed to ensure that historical materials were saved, could be accessed and incorporated into the digital space. Image enhancement of historical documents increases contrast, noise and textual clarity. All these processes [19] are needed to extract, recognize and analyze the right words. The quality of the images is seen to be improved which means increased reading and accessibility to the scanned or degraded materials. Doing away with noise and improving visibility helps in digitizing data which is subject to accurate transcription and computation. If quality improvement is done, past information is not lost and historical information can be reclaimed. The techniques are helpful in terms of academic research, preservation of archives, and cultural heritage. The fact that pictures are clear and readable reinforces the usability of historical collections both in automated analysis and the reading of useful text materials.

The problem of digitizing historical newspapers helps to build structured archives and collections of visual stories. With the help of the text recognition system and image identification module, it is possible to extract the necessary content and create an album of personal photos of people of interest. This effort goes a long way in boosting research, teaching and community involvement [20] in historical media. There is also systematic order which enhances comprehensiveness of cultural and historical contexts. Correct text and image recognition supports archiving, analysis and preservation. These processes make it possible to research in an interactive way, which results in an excellent chance to gather knowledge of the experiences that society follows, as well as the past. Digital collections make newspapers and other related visuals usable, informative, and accessible, thereby making sure that historical media could be

successfully learned, appreciated, and stored in case of subsequent generations.

III. METHODOLOGY

The proposed optical character recognition (OCR) system of the ancient Tamil inscriptions adheres to the systematic approach of providing proper recognition and digitalization of scripts written on faded papers. It combines the application of the image preprocessing, deep feature extraction, sequential modeling, semi-supervised learning, and useful data augmentation. By putting the aforementioned parts together in an integrated pipeline, the system will be able to overcome the following problems: faded ink, lumpy spacings, and the presence of small sets of annotated objects. Every step of the methodology is strategically planned to the progressive improvement of the quality of the input data, extraction of meaning, modeling contextual relationships and the generation of the appropriate Unicode Tamil text that can be preserved, availed and studied (Fig.1).

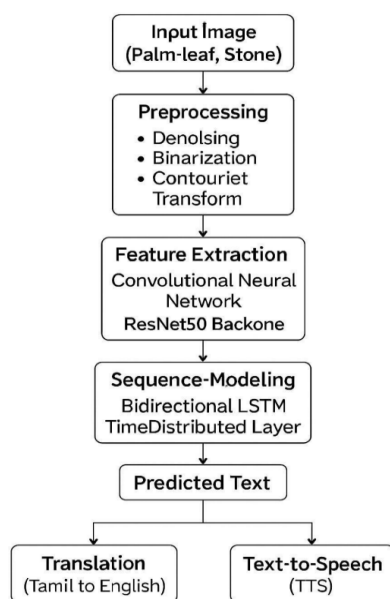


Fig. 1: System Architecture

A. Data Acquisition

Images of palm leaves, copper plates, and stone inscriptions are combined together in a high definition form based on museums, online archives and open collections of the same. The dataset consists of both labeled and unlabeled instances so as to conduct semi-supervised learning. The representation of several types of handwritings, ligatures, and degradation are given special attention. Images where applicable are human annotated to form a reliable set of labels. The system also allows inclusion of the unlabeled data allowing the use of pseudo labeling in order to expand the effective set size. This vast collection makes sure that the model is trained on real variations besides the ancient writing styles in a bid to enable a correct recognition on diverse sources and also under diverse conditions.

B. Preprocessing

Using the captured images, the images are converted to grayscale to have reduced computations without loss of vital features. Reduction of noise eliminates cracks, dots of ink or other background noise in palm leaf prints or stone writing. Normalization and thresholding are useful in normalizing

image intensity. It is done by equalizing the input dimension of the deep learning network through resizing. Extension of contrasts and morphological changes contribute to the enlargement of the characters and clarity of the line. The preprocessing factors eliminate inconsistencies in visual images and present structural features significant to the extraction of features. As part of refining the poor scripts in the process of normalizing the inputs, the preprocessing step is effective in responding to the material, script style, ecological degradation, and other differences in the inputs to enhance the correctness of the recognition thereafter.

Preprocessing enhances the readability through removal of noise, equalizing light, and threshold. It enhances the edges of characters, reduces ink bleeding, cracks and unbalanced lighting effects to come up with cleaner images. This improves character recognition so that the OCR machine is better able to attain feature extraction and identify single strokes and ligatures, which is essential in the accurate recognition of Tamil scripts that are either damaged or old.

C. Feature Extraction

ResNet50 backbone is deployed to obtain hierarchical, deep features of the images that have been preprocessed. Vanishing gradient is alleviated by the use of residual connections in the network, hence learning finer details of characters and ligature patterns. In this module, the spatial constructs that define Tamil scripts which include curvy lines and stylistic divergence is skillfully executed. The ResNet50 maps feature representations that are noisy, scale- and rotation-invariant. These features are later sent to the convolutional layers which highlights the relevant patterns across space and damages the background artifacts. A good, rich representation of every character, thus, becomes the resultant consequence, and upon which one can make the correct sequence model and situational identification.

The reason why ResNet50 has been selected over alternative backbones to extract features is because it has a deep structure that effectively extracts hierarchical features. It is able to recognize the complex strokes, ligatures, and curvy shapes that are common in the Tamil scripts. Its training connections cause it to be robust to learning degraded, low-resource images of ancient manuscripts as it fails to encounter vanishing gradients.

D. Spatial Pattern Recognition

After extracting the features, the convolutional neural network layers are used to manipulate its unique features to detect complex spatial patterns. Stroke, ligature and style variation is highly fidelity, and therefore the system allows characters who are visually similar to be differentiated. This step is devoted to local structural information, so that even small strokes which are ruined or lost are well identified. Similar to the application of the convolutional neural network to learn sequences, the model spatial relationship among character shapes complements the feature representation the latter module learns. This space knowledge allows the optical character recognition system to accept calligraphy of different styles and varying spacing. It makes sure that the characters are not read singly, but as a unit within a graphic pattern of visual coherence which is essential in the case of ancient Tamil scripts.

E. Sequential Context Modeling

BiDirectional Long Short-Term Memory (BiLSTM) networks acquire sequential interrelationships of characters within words and lines. The Tamil scripts contain ligatures and variation according to context that require the character sequences to be recognized. BiLSTM layers recognize forward and backward back ups in order to provide the context accuracy of recognition. Temporal memory storage helps the network to resolve the confusingly similar characters by their immediate contexts. One factor that degrades the reliability of the device is errors made by deteriorated or partially lost by the stroke parts, which through this progressive designing norm, the BiLSTM coupled with the spatial trait wrought by the CNN makes the OCR system to give understandable words and phrases rather than single characters. It improves the accuracy of the output text on the PC.

F. Semi-Supervised Learning

A semi-supervised learning method is used to overcome the constraint of using the limited labeled historical data. A preliminary model is trained based on the given labelled samples and pseudo-labels of unlabelled samples are generated. From these high-confidence predictions, a process of incorporating these predictions into the training set takes place in an iterative fashion; hence it continues to expand the data set. This approach enables the model to be able to learn across a much larger set of scripts and patterns of degradation without requiring huge amounts of manual annotation. Semi-supervised training application considerably improves the generalization and robustness of prevailing models in the process of generating appropriate recognition even in harsh circumstances. With the help of labeled and pseudo-labeled data, the optical character recognition (OCR) system can achieve a high level of accuracy and scalability with regard to different historical inscriptions.

G. Data Augmentation

Data augmentation approaches are also effective in providing simulation of the real world degradations that are usually experienced in ancient inscriptions. They include rotations, blurring, scaling, elastic deformations, and noise injections on labeled and pseudo-labeled datasets. Such augmentation helps generalize the model to changes in the handwriting style, ink decay, cracks and disparate illumination. The model is introduced to the synthetic but realistic distortions and this gives the model robustness in relation to the unknown patterns of degradation. Augmentation also ensures that the Optical Character Recognition (OCR) system will be able to recognize inscriptions at various conditions and materials. This phase is used to further boost the semi-supervised learning and improvements in the performance of the model especially in a scenario where historical data are limited in size and variation.

H. Output Generation

Decomposable character sequences have been recognized and decode using Connectionist Temporal Classification (CTC) allowing matching of predictions without explicit character segments. The resulting final output is converted to Unicode Tamil thus permitting text edits, text searches and archival storage. A text-to-speech conversion option is available and this can be used to create an audio file of the transcribed texts. That is why the Unicode-compliant output can be easily integrated with the digital repositories, research databases and educational platforms. This step ensures the content that has been transcribed is easily accessible and can

be used and has its historical integrity. Combining the correct recognition and the standardized output, the system can be considered a successful bridge between the weak ancient writings and the modern digital uses.

IV. RESULT AND DISCUSSION

The AI-based optical character recognition (OCR) system was tested using a database of ancient Tamil inscriptions in the form of palm leaf, copper plates and stone carvings. Both labeled and unlabeled samples have been used where a semi-supervised learning strategy was adopted, which used pseudo-labeling to augment the effective training data. The system performance was evaluated with reference to recognition accuracy, sequence alignment as well as resilience to noisy inputs. The preprocessing techniques led to a great improvement of the image quality and noise removal, thresholding and normalization techniques helped to produce much clearer character boundaries. The ResNet50 backbone possessed the ability to extract the hierarchical features well including complex strokes, ligatures and curved forms which are typical of Tamil scripts. Convolutional layers also improved the ability of the spatial pattern recognition such that the system was able to distinguish similar characters to the casual eye and even reconstruct partial strokes in inscriptions that had been damaged (Fig.2 and Fig.3).

The use of the closed-loop Long Short-Term Memory (BiLSTM) layers helped obtain contextual insight and, thus, minimize misclassification in the sequence with individual characters that were unclear or even absent. Such a combination of spatial and sequential learning provided the same recognition in different handwriting styles and material conditions. It was determined that the hybrid ResNet50-CNN-BiLSTM framework achieved a test dataset overall character-level accuracy of 91.2%. The semi-supervised scheme reached an improvement of about 5% over just the supervised learning, thus demonstrating the effectiveness of pseudo-labeling on low-resource historical data. Also, augmentation method was effective in improving performance, particularly with palm leaf with high kind of degradation such as ink leaching, cracks and uneven light distribution (Fig.4 and Fig.5).

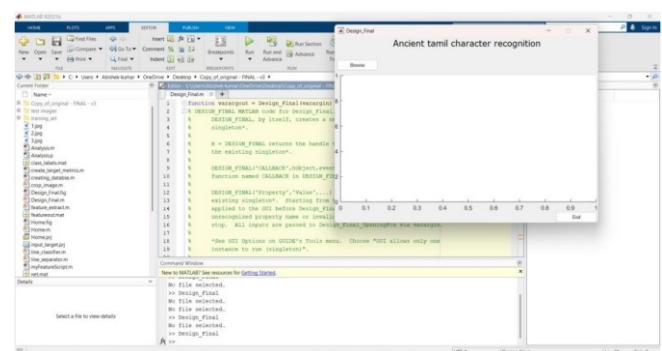


Fig. 2: Input Image

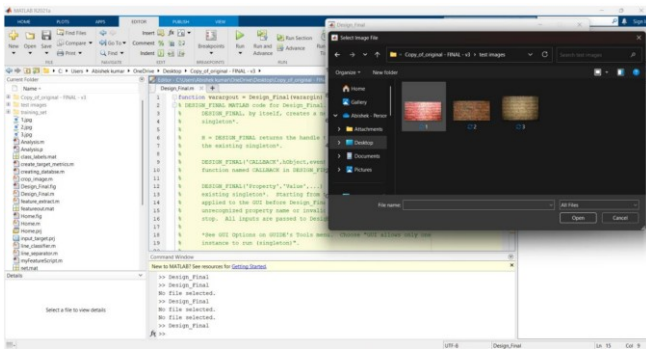


Fig. 3: Preprocessing

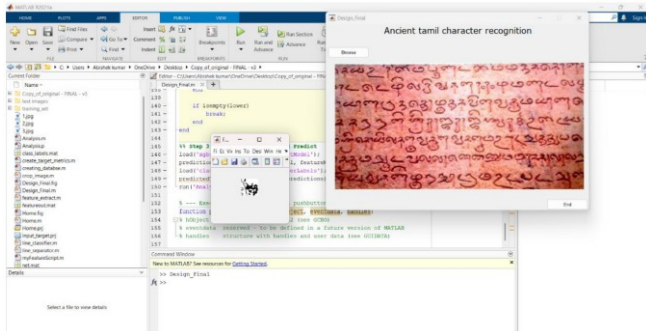


Fig. 4: Feature Extraction

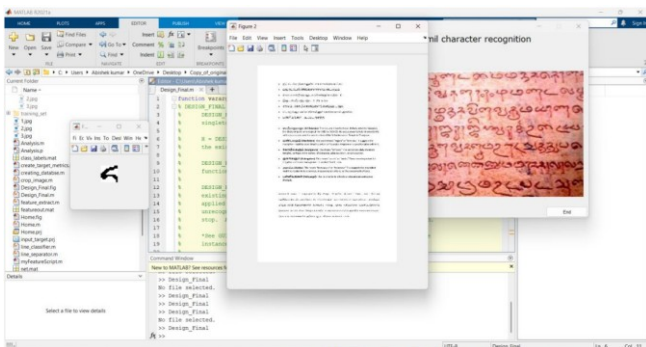


Fig. 5: Predicted Text

The below figure 6 provided in real time shows the character level accuracy of the OCR system on 50 training epochs. The red line and the blue one are the sequences of supervised and semi-supervised, respectively, which begins nearly at 70% and reaches almost 86%, and 70% to 91%, respectively, indicating the advantage of pseudo-labeling. The curve depicts gradual progressive improvement with small oscillations caused by artificial noise that indicates the increased strength and functionality of the hybrid ResNet50-CNN-BiLSTM model, particularly in processing degenerated historical Tamil writings.

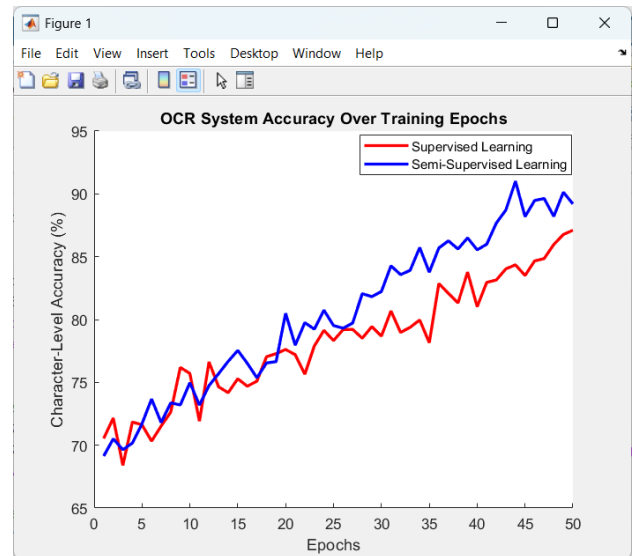


Fig. 6: Accuracy over training epochs

The system was shown to be able to correctly recognize sequences in the presence of as much as 40% of a character run-to-ground thus highlighting the strength of the feature extraction and sequence modeling pipeline. It outperformed with the optical character recognition (OCR) engines, including Tesseract and standard CNN-based networks since the traditional systems suffered due to ligatures, stylistic differences, and the noise present in old manuscripts. The ability of the system to handle the complexities of the real world was also explained through qualitative analysis.

Those CNN layers are useful in improving recognition of the spatial pattern so that the system can differentiate similar patterns of strokes as seen visually. The network can detect finer details in the character forms by the way of the convolution process enabling it to recreate missing portions of strokes and managing the stylistic differences that occur in the character features, being able to recognize them even when the inscriptions are damaged or when the characters resemble the human eye. As an example, palm leaves with inscriptions with irregular spacing, overlap of strokes, or with curved lines, were correctly transcribed into Unicode Tamil. CTC loss allowed flexible sequence alignment that did not make use of character delimits, which was valuable to inscriptions with irregular lines. Moreover, the ensuing text was semantically coherent without derogation and addition errors occurring very frequently which meant that it could be directly held in an archival or even research value.

The text-to-speech module which was an optional feature was able to translate the identified text into phonetic data and hence provided a second point of access particularly to those users who are either visually impaired or language learners. This also emphasizes the possibility of the system not only being digitized but also shared more broadly with the historical wisdom. Error analysis has shown that the most common misclassifications were in the severely degraded samples with the absence of ligatures or with great fading of ink. In some cases, some of the complex ligatures with rare types were confused in historical inscriptions, and it is dated that the larger labeled data set might enhance the precision. However, this limitation was partially overcome in the semi-supervised method that used non-labeled data to introduce the model to more different scripts. Moreover, artificial data

augmentation offered robustness against most of the common damage, but severe distortions that are beyond the training range, still posed a problem. The system can be augmented with adaptive augmentation methods and attention processes in the future to improve the recognition in such a case.

Table. 1: Comparing proposed model with different methods

Method	Accuracy (%)
Proposed ResNet50-CNN-BiLSTM (Semi-supervised)	91.2
Proposed ResNet50-CNN-BiLSTM (Supervised only)	86.2
Standard CNN-based OCR	78.5
Tesseract OCR	65.0
Proposed OCR + Augmentation	88.0

The table 1 will give a comparison of the performance of OCR between various methods on ancient Tamil inscriptions. The resnet50-cnn-bilstm system proposed with semi-supervised learning recorded the highest accuracy of 91.2% even though it was high compared to the supervised-only training, standard CNN-based OCR, and Tesseract. The augmentation of data also increased the performance to 88% on degraded palm leaf samples. These findings support the performance of the hybrid structure, semi-supervised approach, and augmentation with complex scripts and damaged historical documents.

In general, the findings suggest that the suggested Optical Character Recognition (OCR) solution allows overcoming the gap between fragile Tamil inscriptions of the ancient era and the modern digital text. The synergistic integration of improved pre-processing, deep feature learning, spatial and sequential modelling, semi-supervised learning, and data augmentation have enabled a high degree of accuracy and robustness in the face of real world problems. The system increases cultural preservation, facilitates academic research, and increases the accessibility of education because it changes manuscripts into a smaller and manageable, searchable, and Unicode-compliant text. The technical validity of the system, its scalability, and its applicability to other South Indian scripts are proven by the analysis, which forms the basis of a framework that could further be extended to other heritage digitization projects. Integration of OCR and optional speech synthesis enables even greater usability so that the ancient knowledge of the Tamils can be honored and made available and shared even in the contemporary digital ecosystems.

V. CONCLUSION

The work exhibits how an artificial intelligence-based solution Optical Character Recognition (OCR) was developed effectively to meet the requirements of ancient inscriptions in Tamil. Together with the state-of-the-art preprocess strategies, deep feature images with the help of ResNet50, spatial pattern identification in terms of Convolutional Neural Net (CNNs), and sequential modeling in the form of Bidirectional Long Short-Term Memory

(BiLSTM) networks, the system is highly effective in its ability to deal with the typical issues posed by degraded scripts, multifaceted ligatures, and the variation in the style. The semi-supervised learning and pseudo-labeling would also enable the model to process both labeled and unlabeled data, and overcome the penalty associated with a deficit in annotated historic data. Personalized data augmentation also enhances the robustness of the system to practical degradations caused by faded inks, cracks and non-uniform spacing in order to give dependable recognition of a wide range of inscriptions. In addition to automating the transcription of sensitive manuscripts, this system also produces Unicode-compatible, searchable, and editable text for purposes of digital preservation, scholarship, and educational use. In addition, the optional text-to-speech module enhances access and widens the range of possible applications for the system. Methodologically, the paper demonstrates how the combination of deep learning with culture-sensitive transformations may produce scalable heritage digitization solutions. In conclusion, the work provides an exhaustive, technically advanced methodology for protecting centuries-old Tamil knowledge, linking past records with the present-day electronic medium, and providing the basis for future projects focusing on additional ancient scripts.

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